

Global warming drives the evolutionary rate of H1N1 and H3N2 influenza viruses

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Abstract

The H1N1 and H3N2 subtypes of the seasonal influenza A virus have circulated among humans for decades. Because of their pathogenicity, these viruses have been extensively studied from epidemiological, molecular and evolutionary perspectives. Their seasonality is primarily driven by variations in temperature and humidity, which also play a key role in shaping outbreak dynamics. Yet, despite numerous studies estimating the substitution rates of different Influenza A subtypes, it remains unclear whether these rates change over time in response to shifting climate conditions. To address this outstanding question, we collected genomic sequences of the hemagglutinin and neuraminidase genes for both H1N1 and H3N2 subtypes circulating worldwide. Keeping only sequences from countries with records spanning at least 2 decades, we performed a Bayesian analysis to estimate substitution rates. We show that substitution rate is driven by temperature for both subtypes and genes for multiple countries (Australia, China, Japan, Netherland, Russia, Thailand, U.S.A.). When it is not the case, a power analysis indicates a potential lack of sequences to detect the effect. As temperatures keep increasing due to global warming, further research is needed to understand if a speed up in the influenza evolution rate has any impact on epidemic burden.

Keywords: Virus, Evolution, Influenza, Global Warming, Substitution Rate

1. Introduction

Type A influenza is a seasonal virus causing 290,000 to 650,000 deaths worldwide annually [1]. Despite widespread vaccination campaigns, annual Influenza A epidemics persist largely due to antigenic drift. Because negative-sense RNA viruses like influenza A lack a proofreading mechanism, mutations steadily accumulate during each replication cycle. Over time, this ongoing mutation process generates new antigenic variants capable of evading host immunity acquired through vaccination or prior infection [2]. As a result, these variants can reach fixation, either via random chance, also called genetic drift, or through selection, and can be studied through phylogenetic analyses [3–6]. Surprisingly, such studies have not tested whether substitution rates vary through time and if such variations are linked to outside variables. Furthermore, mutation rates (expressed in mutations/site/cycle) are correlated with substitution rates (expressed in substitutions/site/year) in RNA viruses [7]. Thus, factors affecting viral transmissions and the number of viral replication cycles could affect substitution rates.

Temperature is one key environmental factor that could mediate these effects. Influenza exhibits well-known seasonality, with outbreaks peaking in winter in temperate regions and less distinct timing in tropical zones [8]. The seasonality of influenza has been linked to local temperature and humidity conditions, both of which influence viral transmissibility [8–10]. Experimental and epidemiological evidence shows that influenza viruses transmit less efficiently under higher temperature and humidity, suggesting that shifts in global climate could modify epidemic patterns. With global temperatures projected to rise by 0.5 to 2.8°C over the twenty-first century [11], the dynamics of influenza transmission may be entering a period of unprecedented change. Indeed, warming winters in the United States have already been associated with milder influenza seasons, followed by more severe outbreaks the next year, likely due to altered population susceptibility [12].

While the epidemiological impacts of climate change on influenza seasonality are increasingly recognized, its potential influence on viral evolutionary rates remains unexplored. Changes in replication opportunities driven by temperature could modify the rate at which viral genomes evolve, particularly in genes under strong selective pressure. For influenza A, the hemagglutinin (HA) gene encodes the surface protein responsible for cell binding via sialic acid and is under intense immune-driven positive selection [2,13]. The neuraminidase (NA) gene, which facilitates virion release through cleavage of sialic acid residues, also contains antigenic sites subject to selection, though to a lesser extent [2,13].

Past efforts have examined how socioeconomic and environmental indicators, such as healthcare spending, population density, or temperature, correlate with influenza subtype diversity and genomic

sequence diversity [14,15]. However, these studies often lacked formal adjustment for sampling biases and did not focus specifically on substitution rate variation.

In this study, we investigate whether temperature influences the evolutionary rate of Influenza A viruses in the context of global climate change. Using extensive publicly available genomic data, we estimate temporal substitution rates for the HA and NA genes in the H1N1 and H3N2 subtypes across multiple countries and assess their correlation with local temperature trends. This approach provides new insights into how rising global temperatures might influence the molecular evolution and adaptability of influenza viruses in the coming decades.

2. Material and methods

(a) Data acquisition

The genomic sequences coding for the hemagglutinin (HA) and the neuraminidase (NA) proteins for the H1N1 and H3N2 influenza viruses were retrieved from the NCBI Influenza database [16]. On August 18th, 2023, we downloaded a total of 7,559 HA and 15,966 NA sequences for H1N1, and 28,176 HA and 26,088 NA sequences for H3N2. For each variant and coding region separately, we filtered out sequences with unknown dates of collections and plotted the distribution of sequences according to years of collection and countries of origin. To ensure reliable estimates of evolutionary rates over a sufficiently long period, we excluded countries with sequences available for fewer than 10 distinct years and with less than 20 years of records. For countries meeting these criteria, we additionally downloaded all available complete sequences for both subtypes and genes from the GISAID database [17] to maximize sample size.

(b) Preprocessing

All sequences were grouped by country, year, and subtype and gene. For each combination, sequences were subsampled using CD-HIT [18] to extract representative sequences and to remove redundant ones, thus reducing the dataset size and limiting the computational burden of downstream analyses. To test the impact of this data-reduction step, CD-HIT was used at two thresholds (99% and 99.7%), effectively leading us to doubling the number of datasets for each combination of country/year/subtype/gene. As extracting representative sequences can inflate diversity and hence increase rate estimates, we created two additional and distinct datasets where the sequences were randomly subsampled (*contra* using CD-HIT),

drawing for each combination of country/year/subtype/gene an equal number of sequences as those obtained with CD-HIT. All four data collection schemes were pre-processed and analysed as described below.

We aligned sequences by subtype, gene and country using MAFFT ver.7.471 [19], and trimmed the alignments with trimAl ver.1.5.rev0 [20] using the "gappyout" and "keepheader" arguments. We then manually checked each alignment for each one to start with a methionine and end with a stop codon. Few alignments were incorrectly trimmed by trimAl, leading to an out-of-frame alignment. In such cases, we manually trimmed the alignments, deleting positions containing sequencing errors that manifested when less than 5% of the sequences contained nucleotides at a position, creating a gap in every other sequence. After trimming, we removed sequences containing 20 or more gaps/unidentified nucleotides in a row to only keep high quality sequences.

(c) Analyses

We performed a Bayesian analysis using Markov Chain Monte Carlo (MCMC) samplers to reconstruct phylogenies using the BEAST software ver.1.10.4 [21]. For each combination of gene, variant, and country, and for all 4 data collection schemes, we ran each MCMC sampler twice independently under an uncorrelated relaxed molecular clock model [22] and a General Time Reversible (GTR) substitution model [23] with Γ distributed [24] rates across sites. Each sampler ran until they converged with an Effective Sample Size of at least 175 or for a maximum of 100 million steps. Convergence was visually assessed by plotting the joint distribution according to the steps in the MCMC chain for each pair of analyses after conservatively removing half of the steps as burnin. We combined each pair of runs with logcombiner [25], and used treeannotator [25] to summarise the results. Finally, we extracted substitution rates for each branch of each tree and matched them to the date of each of their child node using the Treeio R ver.3.21 package [26] in R ver.4.4.1[27] .

To assess if substitution rates are correlated with time, we ran robust linear regressions for each analysis between substitution rates and the dates of the associated child node of the branch using the "robust" R package ver.0.7-4 [27,28]. Furthermore, to test if the substitution rates were correlated with global warming, we collected the yearly average temperature for each country from the Climatic Research Unit (CRU) year-by-year variation of selected climate variables by country (CY) ver.4.08 dataset, which is derived from the CRU gridded Time Series (TS) ver.4.06 dataset [29]. For each country in our datasets, we built a robust MM linear regression model to estimate the average annual temperature over time. Then, we used those estimates to match the substitution rate of each branch to the average temperature

over the branch period. We then ran a robust linear regression between substitution rates and the estimated average temperatures as described above.

Finally, we tested if genes and subtypes increase at a similar rate for countries with a significant correlation between rates and temperatures. For this, we averaged the substitution rates of analysis with a significant correlation for each year and performed a robust ANCOVA with time as a covariate and subtype/coding region as independent groups. We used the MASS ver.3.60.2 and the emmeans packages ver.1.11.2.8 to obtain multiple comparisons between groups with p -values adjusted with Tukey's method.

3. Results

After performing all preprocessing steps, H1N1 HA and NA genomic sequences from five different countries were retained while sequences from ten and 13 countries were kept for H3N2 HA and H3N2 NA, respectively (Table S1). H3N2 datasets had 1.5 to 2.3 times more sequences than H1N1 datasets, and increasing the threshold from 99% to 99.7% increased the number of sequences up to 7 times (Table S1). However, the sequences were not evenly distributed amongst countries, as sequence counts were dominated by the U.S.A., Australia, Japan, and China (Table S2). Overall, 19 analyses revealed a significant ($p < 0.01$) correlation between substitution rates and time (Figure S1). All significant correlation were positive, indicating that the substitution rate for some genes of the influenza strain increased over time in some countries. This temporal trend is observed across seven countries, spanning both hemispheres, including the U.S.A., Russia, Japan, Netherlands, China, Australia and Chile). As expected, the CD-HIT analyses at a high similarity threshold returned the largest number of significant results (Figure 1), with eleven and four significant results at the 99.7% and 99% similarity thresholds, respectively, while under random sampling these numbers fall to three and one.

Given the ongoing increase in annual temperatures attributable to global warming, we hypothesised that the analyses that significantly correlated with time would also demonstrate significant correlations with yearly temperature estimates. The results, shown in Figure 1, support this hypothesis. Specifically, 18 out of the 19 analyses exhibiting significant temporal correlations remained significant when correlated with temperature data. Notably, the H1N1 Neuraminidase analysis from the U.S.A., employing a 99% threshold via random sampling, lost significance, whereas the H3N2 Hemagglutinin analysis from Thailand, using CD-HIT with a 99.7% similarity threshold, became significant. To assess whether the number of sequences influenced the ability to detect correlations between substitution rates and temperatures, we added regression lines in Figure 1 to examine the relationship between sequence count significance. The results indicate no significant correlation for the H1N1 subtype, but a significant

negative correlation ($p < 0.01$) was observed for H3N2, suggesting that, for this subtype, the effect could be detected in more countries if more sequences were available. Despite these findings, the correlations are weak, with R^2 values ranging from less than 1% to 4%, most clustering between 1 and 3% (Table 1). The slopes of these regressions ranged from 2.66×10^{-6} to 5.25×10^{-5} substitutions per site per year, representing marginal increases relative to the average substitution rate of approximately 1×10^{-3} substitutions per site per year.

Furthermore, we tested whether the subtype or the gene segment evolved at different rates. Robust ANCOVA results (Figure 2) revealed significant differences in mean substitution rates between subtypes and genes for both sampling methods. Under CD-HIT, no significant interaction between time and subtype or gene (i.e., no difference in slope) was detected. However, H1N1 HA, H1N1 NA, and H3N2 HA presented significantly higher mean substitution rates (differences of $\sim 10^{-4}$ substitutions/site/year) than H3N2 NA (Table S3). For analyses using the random sampling method, a significant interaction was discovered between H1N1 HA and H3N2 NA, indicating that the substitution rate of H1N1 HA increases faster (Table S4). Additionally, differences in mean rates were also significant: H1N1 HA exceeded H1N1 NA and H3N2 NA, while H1N1 NA was lower than H3N2 NA, with overall differences in rates on the order of 10^{-4} substitutions per site per year (Table S4).

4. Discussion

Although numerous studies have estimated evolution rates of influenza A viruses [13,14,30–32], none have investigated how these rates change over extended periods. Our findings (Figure 1) reveal an increase in nucleotide substitution rates over time within the hemagglutinin gene (HA) of the H1N1 virus in the U.S.A., Russia, and Japan. A similar increase is found for the neuraminidase gene (NA) in the U.S.A. and Russia. This pattern is detected in a broader range of countries for the H3N2 subtype, likely reflecting the availability of a larger sequence dataset. Specifically, substitution rate increases are detected in Australia, China, and the Netherlands for the HA, as well as in Chile and Russia for NA. For all the analyses exhibiting a significant increase, the rise is marginal (increase of $\sim 10^{-5}$ substitutions/site/year) relative to the mean substitution rates ($\sim 10^{-3}$ substitutions/site/year) consistent with published estimates for Influenza A [13,14,30–32]. The coefficient of determination (R^2) for each analysis was low, explaining only a small fraction of the variation in substitution rates, consistent with a minor yet persistent temporal effect.

Subsampling methods impacted the detection of significant results, with 44% of analyses using CD-HIT clustering at 99.7% similarity showing significance ($p < 0.01$), compared to 18% for random

sampling at the same threshold. At the 99% threshold, CD-HIT and random sampling resulted in 9% and 0% of the analyses reaching significance, respectively. Random sampling, however, does not prioritize sequence diversity or similarity, hence leading to datasets that can include redundant or highly similar sequences. This redundancy led to a significant decrease in estimates of substitution rates at 99.7% similarity, but this was not the case for the 99% threshold (Table S5), possibly due to a reduced amount of available sequences. A decrease in substitution rates reduces statistical power, and thereby decreases the likelihood of detecting significant correlations. Thus, CD-HIT's approach of lowering redundancy through clustering enhances the signal-to-noise ratio, leads to significant patterns more apparent and statistically detectable. These methodological differences likely underlie the divergent ANCOVA results we obtained, as optimising genetic diversity representation provides a more robust estimate of substitution rate dynamics. ANCOVA results for CD-HIT-sampled data (Figure 2) revealed no significant interaction between time and subtype/gene, indicating similar substitution rate increases across H1N1 and H3N2 subtypes and gene segments.

The exact mechanism driving this subtle increase in substitution rate remains unclear but may involve several evolutionary processes. Mutation rates closely track substitution rates in RNA viruses [7], with mutation rates being subject to respond to selection [33]. Additionally, RNA viruses encode their own replication machinery, enabling selective optimisation of mutation rates to increase fitness [33]. Elevated mutation rates can enhance viruses adaptability by generating beneficial mutations [33,34], but only up to a threshold beyond which deleterious mutations accumulate, as demonstrated by lethal mutagenesis experiments in poliovirus and influenza [35,36]. RNA viruses may have evolved just under this lethal threshold, as selection for genetic diversity always pushes mutation rates closer to this limit. This could explain the gradual increase in substitution rates observed in H1N1 and H3N2, but assessing whether RNA viruses have reached their optimal mutation rates due to selection has been challenging [33,37].

Transmission bottlenecks may also contribute to the small increase in substitution rate. Indeed, warmer winters often lead to milder influenza seasons, with reduced infection rates, followed by more severe epidemics [12]. Such bottlenecks reduce the effective population size and can relaxing purifying selection, hence accelerating substitution rates [7,38].

Finally, increased genomic diversity in the datasets we analysed may drive the observed substitution rate rise. To our knowledge, changes in genomic diversity over time for H1N1 and H3N2 subtypes have not been formally investigated. However, globalisation and the rise of international travel, which more than doubled between 2000 and 2020 [39], could be contributing to the rapid dissemination of novel variants throughout the world, as highlighted during the 2020 SARS-CoV-2 pandemic [40].

Thus, an increase in viral diversity caused by viral spread through international travel in recent years could have led to increased substitution rates in our analyses: a hypothesis that still remains to be tested.

5. Conclusion

Although the effect detected in this study is modest, the evolutionary trajectory of influenza viruses remains uncertain as the global temperatures continue to rise, and the impact of an increase in substitution rate on the fitness of the H1N1 and H3N2 subtypes remains to be clarified. If repeated viral population bottlenecks drive the increase in substitution rate, as we here found, many of the new mutations fixed in the population could be synonymous, neutral, or slightly deleterious, and thus unlikely to enhance viral fitness. Consequently, the evolutionary and epidemiological implications of this subtle rate increase remain unclear. Further research is warranted to elucidate the long-term dynamics of substitution rate changes and to assess their potential effects on viral fitness and epidemic severity.

Data accessibility

The accessions numbers for the sequences and the exact parameters used in this study are available as xml files along with a csv file containing the results of our analysis. The associated code is available at <https://github.com/sarisbro>

Declaration of AI use

No AI-assisted technology was used in this work.

Authors' contributions

M.V. : data curation, formal analysis, investigation, methodology, validation, writing—original draft, writing—review and editing; R.M.,: initial data curation, formal analysis, writing—original draft, writing—review and editing; S.A.-B.: conceptualization, investigation, resources, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Funding

This work was funded by the Natural Sciences and Engineering Research Council of Canada (NSERC), the Ontario Graduate Scholarship (OGS) and by the University of Ottawa.

Acknowledgements

We gratefully acknowledge all GISAID data contributors, i.e., the Authors and their Originating laboratories responsible for obtaining the specimens, and their Submitting laboratories for generating the genetic sequence and metadata and sharing via the GISAID Initiative, on which this research is based. Additionally, the authors thank the Digital Research Alliance of Canada for providing us with access to their high-performance clusters.

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Table 1: Summary table for the analysis with a significant correlation between substitution rates and time.

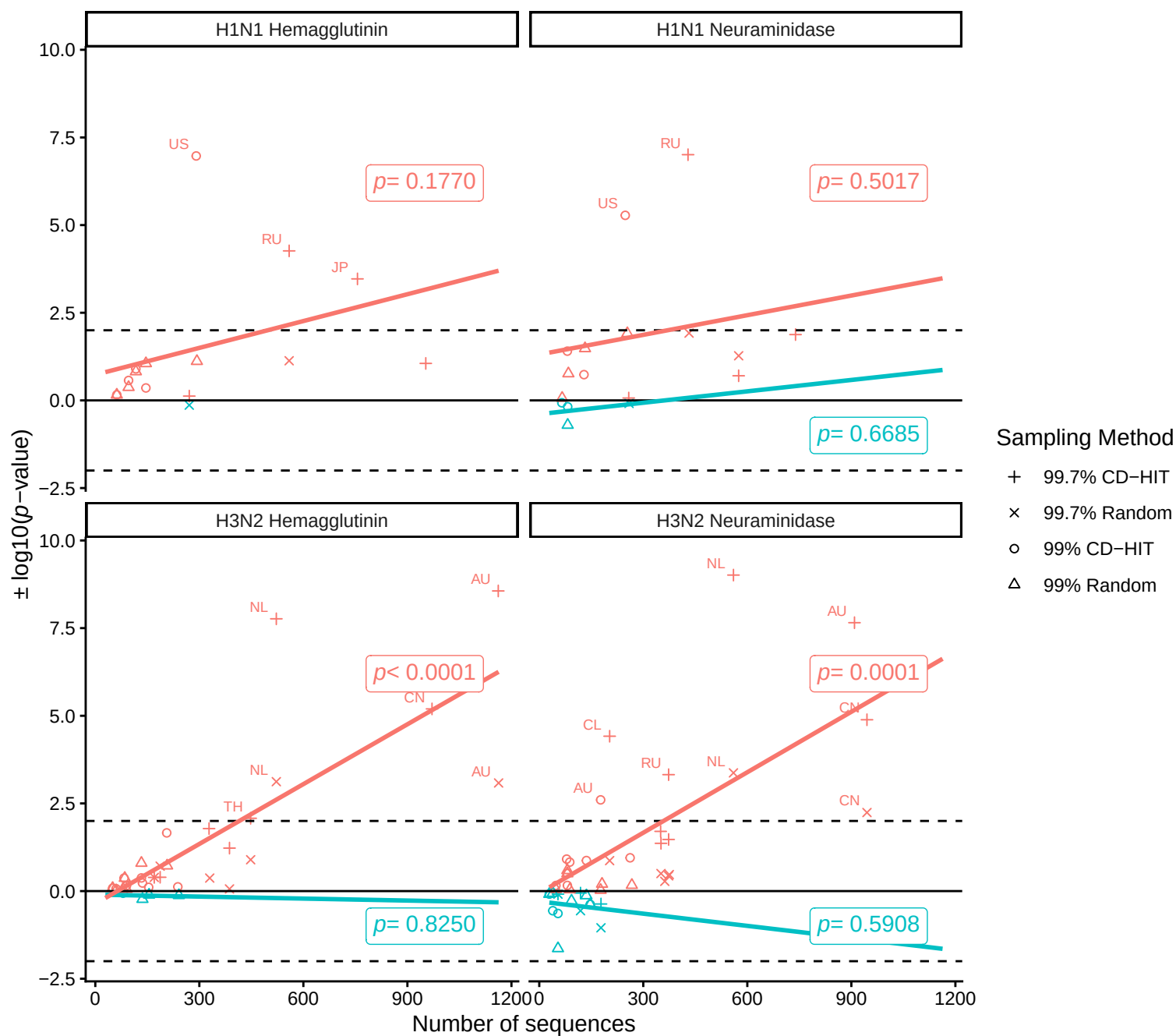
Country	Subtype and gene	Sampling	Slope	Adjusted <i>p</i> -value	R ²	Mean Substitution rate
Japan	H1N1 HA	99.7% CD-HIT	2.37x10 ⁻⁵	0.0007	0.007	5.74x10 ⁻³
Russia	H1N1 HA	99.7% CD-HIT	3.02x10 ⁻⁵	0.0004	0.012	4.80x10 ⁻³
U.S.A.	H1N1 HA	99.7% CD-HIT	1.47x10 ⁻⁵	< 0.0001	0.036	5.88x10 ⁻³
Russia	H1N1 NA	99.7% CD-HIT	5.25x10 ⁻⁵	< 0.0001	0.029	5.26x10 ⁻³
U.S.A.	H1N1 NA	99% CD-HIT	1.21x10 ⁻⁵	< 0.0001	0.042	3.13x10 ⁻³
U.S.A.	H1N1 NA	99% Random	4.01x10 ⁻⁶	0.0097	0.009	2.85x10 ⁻³
Australia	H3N2 HA	99.7% CD-HIT	2.74x10 ⁻⁵	< 0.0001	0.014	5.16x10 ⁻³
Australia	H3N2 HA	99.7% Random	1.28x10 ⁻⁵	0.0009	0.003	4.45x10 ⁻³
China	H3N2 HA	99.7% CD-HIT	2.53x10 ⁻⁵	< 0.0001	0.010	4.16x10 ⁻³
Netherlands	H3N2 HA	99.7% CD-HIT	1.18x10 ⁻⁵	< 0.0001	0.026	4.57x10 ⁻³
Netherlands	H3N2 HA	99.7% Random	4.68x10 ⁻⁶	0.0010	0.007	4.34x10 ⁻³
Australia	H3N2 NA	99.7% CD-HIT	1.93x10 ⁻⁵	< 0.0001	0.013	4.34x10 ⁻³
Australia	H3N2 NA	99% CD-HIT	1.63x10 ⁻⁵	0.0031	0.019	3.19x10 ⁻³
Chile	H3N2 NA	99.7% CD-HIT	2.16x10 ⁻⁵	0.0001	0.030	4.16x10 ⁻³
China	H3N2 NA	99.7% CD-HIT	1.60x10 ⁻⁵	< 0.0001	0.010	3.12x10 ⁻³
China	H3N2 NA	99.7% Random	7.31x10 ⁻⁶	0.0055	0.004	2.92x10 ⁻³
Netherlands	H3N2 NA	99.7% CD-HIT	9.61x10 ⁻⁶	< 0.0001	0.032	3.99x10 ⁻³
Netherlands	H3N2 NA	99.7% Random	2.66x10 ⁻⁶	0.0003	0.010	3.60x10 ⁻³
Russia	H3N2 NA	99.7% CD-HIT	1.17x10 ⁻⁵	0.0006	0.016	4.16x10 ⁻³

HA: Hemagglutinin; NA: Neuraminidase.

Figure captions

Figure 1: Relationship between the p -value and the number of sequences for each analysis when correlating substitution rates and yearly temperature. Analysis in red showed a positive correlation between substitution rates and time while analysis in blue showed a negative correlation. The dash lines represent the significance threshold for $p < 0.01$. The name of the countries with a significant correlation are displayed using the ISO 3166-1 alpha-2 notation.

Figure 2: Yearly substitution rate averaged across countries displaying a significant correlation between rates and temperatures. The average substitution rates were separated by sampling method and subtypes / coding regions. Robust ANCOVAs between substitution rates and subtypes with time as a covariate for the 99.7 CD-HIT and Random sampling methods were performed. A one-on-one post-hoc comparison test adjusted for multiple comparison was carried out to test differences in substitution rate for each subtype/coding region. Stars in red indicate an ANCOVA where there is a significant interaction between time and subtype/coding region while grey stars indicate that the interaction term was not significant. Stars indicate significance with $p < 0.01$ and with $p < 0.0001$, respectively.



Average Substitution Rate (substitution/site/year)

